

Ben Sedran

Mechanical Engineering and Computer Science

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Expected May 2028

Bachelor of Science in Mechanical Engineering, Computer Science Minor **GPA: 3.75**

MECHANICAL ENGINEERING COURSES:

- **Statics and Mechanics of Solids:** Stress, deflection, and failure analysis of trusses, frames, beams, shafts, and columns and solved statically indeterminate problems involving various loading modes in 2 and 3 dimensions
- **Thermodynamics:** Made and justified assumptions about thermodynamic systems, cycles, and devices in order to analyze and make calculations of efficiency, back work ratio, and output states

COMPUTER SCIENCE COURSES:

- **Object Oriented Programming and Data Structures:** Implemented with proper documentation abstract data types using java, coded a formula evaluator for CSV files along with unit and black box tests, and used Dijkstra's Algorithm to create an interactive image selector with a GUI
- **Mathematics of Computing:** Wrote formal proofs including induction and graph induction, combinatoric proofs and set cardinality proofs, probability and Bayes' Rule and random variables, and finite automata

CURRENT COURSEWORK (Spring 2026):

- Dynamics
- Mechanical Design
- Computer System Organization
- Data Structures and Functional Programming

WORK EXPERIENCE

Billerica Boys and Girls Club, Billerica, MA, *Pool Supervisor and Swim Instructor* **July 2021-Aug 2025**

- Taught swim lessons to children aged 3-12 in both individual and group classes year round
- Was promoted to supervisor in 2024, gave swim tests and used salesforce to enroll and build classes
- Trained new instructors on safety in their classes and how to effectively teach
- Measured and maintained correct chemical concentrations in the pool

Spring Brook Park, Bedford, MA, *Lifeguard and Swim Instructor*

June 2022-Aug 2025

- Set up and closed the park every summer, included building and deconstructing a floating dock
- Taught basic to intermediate swimming skills to the town summer camp and lifeguarded afternoons
- Made several saves of struggling swimmers and enacted our missing child search protocol

SPECIALIZED SKILLS

Programming languages: Java, C, OCaml

Programs: Autodesk Fusion 360, MATLAB, LaTeX

Certifications: Machine Shop Certified

Languages: Spanish (intermediate)

ACTIVITIES AND INTERESTS

- Member of the Cornell Club Water Polo team
- Rock climber and mountain climber
- Avid chess player with an 1800+ rating on chess.com
- Woodworker who has built furniture for my apartment